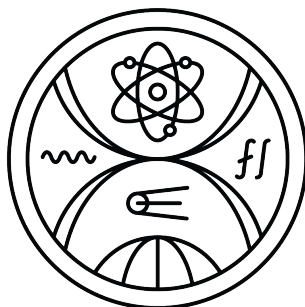


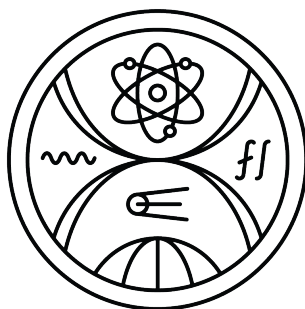
UNIVERZITA KOMENSKÉHO V BRATISLAVE
FAKULTA MATEMATIKY, FYZIKY A INFORMATIKY



AKTÍVNA PERCEPCIA V ROBOTICKOM VIDENÍ

Diplomová práca

UNIVERZITA KOMENSKÉHO V BRATISLAVE
FAKULTA MATEMATIKY, FYZIKY A INFORMATIKY



AKTÍVNA PERCEPCIA V ROBOTICKOM VIDENÍ

Diplomová práca

Študijný program: Aplikovaná informatika
Študijný odbor: Aplikovaná informatika
Školiace pracovisko: Katedra aplikovanej informatiky
Vedúci práce: prof. Ing. Igor Farkaš, Dr.



Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta: Bc. Patrik Modrovský
Študijný program: aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor: informatika
Typ záverečnej práce: diplomová
Jazyk záverečnej práce: slovenský
Sekundárny jazyk: anglický

Názov: Aktívna percepcia v robotickom videní
Active perception in robotic vision

Anotácia: Proces percepcie v robotickom kontexte je charakteristický aktívnou interakciou s objektmi počas učenia. Tieto procesy súčasne umožňujú "lacné" generovanie rozmanitých tréningových dát, ktoré sa dajú využiť na rôzne úlohy, napríklad robustne rozpoznávať objekty.

Cieľ: 1. Štúdium literatúry v oblasti aktívnej percepcie.
2. Návrh modelu neurónovej siete ktorá umožní robotovi skúmať objekt držaný v ruke v rôznych uhloch.
3. Vyhodnotenie funkčnosti a úspešnosti modelu a návrh možných vylepšení.

Literatúra: End to end active perception, I. Kostrikov, D. Erhan & S. Levine, NIPS 2016, http://www.dumitru.ca/files/DL_symposium_active_perception.pdf
Active Perception and Representation for Robotic Manipulation, Y.Zaky, G. Paruthi, B. Tripp, J. Bergstra, <https://arxiv.org/abs/2003.06734>

Vedúci: prof. Ing. Igor Farkaš, Dr.
Katedra: FMFI.KAI - Katedra aplikovanej informatiky
Vedúci katedry: doc. RNDr. Tatiana Jajcayová, PhD.
Dátum zadania: 16.10.2022
Dátum schválenia: 06.11.2022

prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

.....
študent

.....
vedúci práce

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

Bratislava, 2023

.....
Bc. Patrik Modrovský

Acknowledgement

do later

Abstract

later

Keywords: robots

Abstrakt

later

Klíčové slová: roboti

Obsah

1	Introduction	2
1.1	Neural networks	2
1.1.1	History	2
1.1.2	Different models of neural networks	3
1.2	Active vision	5

Motivation

nic moc zatiaľ

Kapitola 1

Introduction

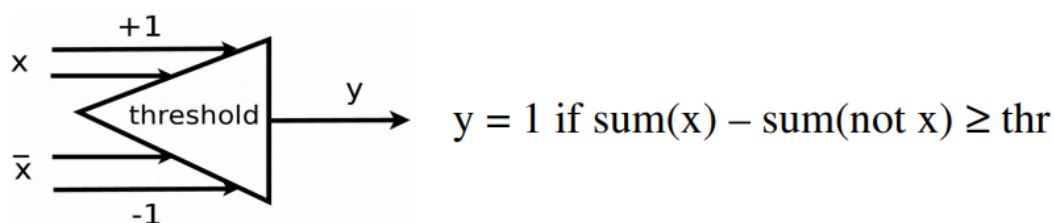
1.1 Neural networks

An Artificial Neural Network (ANN) is a processing system made from large number of interconnected simple processing elements - neurons. Architecture ANNs was inspired by structure of biological brain - both brain and ANN process information similarly and learns from examples.

This creates different approach to problem solving compared to conventional computing. Instead of predictable algorithmic approach, ANNs allows solving problems without understanding of problem, but with lower accuracy. [3]

1.1.1 History

In 1943 neurophysiologist Warren McCulloch and mathematician, Walter Pitts developed first model of ANN. Neurons in their models had fixed threshold and were able to simulate simple logic functions. [3]



Obr. 1.1: Model of an artificial neuron according to McCulloch and Pitts [1]

After initial period of enthusiasm and few other contributions to the field, period of frustration followed. During that time there were only few researchers continued work on solving problems with neural networks. During this time Klopff developed basis for learning and Werbos developed back-propagation algorithm.

This period lasted up to the 1980s, when multiple new contributions renewed wider interest. Hopfield described recurrent neural networks and presented paper *Neural Networks and Physical Systems with Emergent Collective Computational Abilities*. [3]

Neural networks become recently much more publicly known, caused by advances in computing power allowing much bigger number of neurons. It allowed for much more complex tasks as image generation from natural language prompts or chatbots capable of deeper conversations.

1.1.2 Different models of neural networks

There are hundreds of different models and architectures of neural networks with different advantages and disadvantages for different purposes. Some of the more basic are:

- Single perceptrons
- Multi-layer perceptrons
- Recurrent neural network
- Self-Organizing Maps
- Radial Basis Function Network
- Hopfield autoassociative memory

In practise, multiple architectures are often combined to achieve desired outcome. [1]

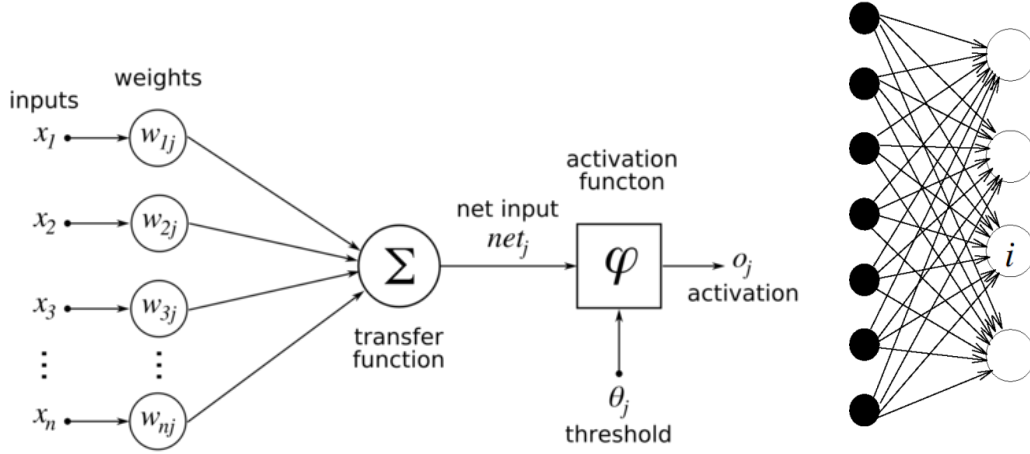
Simple perceptron

It was proposed in 1958 by American psychologist, F. Rosenblatt. It is more general model, with free parameters, stochastic connectivity and threshold elements. [1]

Activation of this perceptron consists of weighted sum of inputs, from which we subtract threshold θ (1.1). Result is then run through activation function, either uni/bipolar for discrete perceptron or sigmoid for continuous perceptron. In practise, threshold is expressed as another weighted input x_{n+1} , with static value of -1 (1.2).

$$o_j = f\left(\sum_{i=1}^n w_{ij}x_i - \theta_j\right) \quad (1.1)$$

$$o_j = f\left(\sum_{i=1}^{n+1} w_{ij}x_i\right) \quad (1.2)$$



Obr. 1.2: Model of simple perceptron and layer of simple perceptrons [1]

Learning rule for adjust weights in discrete perceptron:

$$w_j(t+1) = w_j(t) + \alpha(d - y)x_j \quad (1.3)$$

And for continuous perceptrons:

$$w_j(t+1) = w_j(t) + \alpha(d^{(p)} - y^{(p)})f'x_j \quad (1.4)$$

Simple perceptron can solve only lineary separable classes. This caused loss of interest in neural networks in many researchers 1970s as many real problems are lineary non-separable, thus simple perceptron cannot be used to solve them.

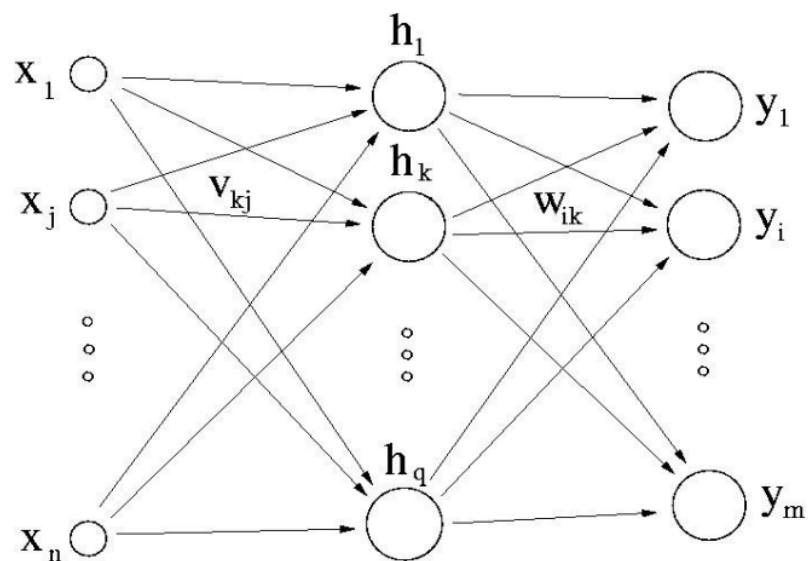
Multi-layer perceptron

Probably one of most used architectures today. Contains additional hidden layers and is generalisation of simple perceptrons. Originates from *Rumelhart & McClelland: Parallel distributed processing* but was earlier described by Werbos in 1974. It is response to critique on perceptrons.[1]

Activation function....

$$h_k = f\left(\sum_{j=1}^n v_{kj}x_j - \theta\right) \quad (1.5)$$

$$y_i = f\left(\sum_{j=1}^{q+1} w_{ik}h_k\right) \quad (1.6)$$



Obr. 1.3: Model of multi-layer neural network [1]

1.2 Active vision

TEST 2

Literatúra

- [1] Igor Farkaš. Neural networks. page 192. Knížničné a edičné centrum FMFI UK, 2011.
- [2] Dominique A. Heger. An introduction to artificial neural networks (ann)-methods , abstraction , and usage. 2015.
- [3] Bohdan Macukow. Neural networks – state of art, brief history, basic models and architecture. In Khalid Saeed and Władysław Homenda, editors, *Computer Information Systems and Industrial Management*, pages 3–14, Cham, 2016. Springer International Publishing.
- [4] Hai Nguyen and Hung La. Review of deep reinforcement learning for robot manipulation. In *2019 Third IEEE International Conference on Robotic Computing (IRC)*, pages 590–595, 2019.
- [5] Youssef Zaky, Gaurav Paruthi, Bryan Tripp, and James Bergstra. Active perception and representation for robotic manipulation. *CoRR*, abs/2003.06734, 2020.